

Uniquely covered induced paths in triangle-free graphs: a negative answer to a question of Scott and Seymour

github.com/graph-theory-AI

Machine-generated note (see provenance box)

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Provenance

The mathematics in this note was produced without human intervention, in three stages. (1) The construction and proof were found by the language model `gpt-5.6-so1` in a single autonomous pass on 2026-09-01, as part of a large sweep over open problems catalogued from arXiv (catalog id 1702.01094__01; original writeup in `attacks/1702.01094__01/output.md`). (2) An independent adversarial referee agent (`claude-fable-5`) re-derived every step, confirmed the question verbatim against the TeX source of the paper, and verified the deterministic core ([Lemma 2.1](#)) by exhaustive enumeration on eight graphs, including a 100-vertex subdivided Petersen graph, as well as every inequality of the probabilistic part numerically; its verdict was **CONFIRMED**. Its report is reproduced verbatim in [Appendix B](#). (3) On 2026-09-09 the writeup was rewritten into the present note by `claude-fable-5.1`: the text was reorganised with full proofs in [Appendix A](#), and the referee's remarks on interpretation and on what remains open were added as [Section 4](#). No mathematical content was changed. **No human mathematician has checked this argument.** We circulate it to obtain exactly that: is the proof correct, and is the result new?

Abstract

Scott and Seymour asked whether, for a triangle-free graph G of very large chromatic number and an arbitrary family $\mathcal{A}$ of stable sets covering $V(G)$ (not necessarily pairwise disjoint), there must be an s -vertex induced path P each of whose vertices v is *uniquely covered*, i.e. $X \cap V(P) = \{v\}$ for some $X \in \mathcal{A}$. The partition case is their rainbow-path theorem. We show that the general question has a negative answer for every $s \geq 3$: for every C there is a connected triangle-free graph G with $\chi(G) > C$ and a stable-set cover $\mathcal{A}$ of $V(G)$ such that no path of G with at least three vertices, induced or not, has all its vertices uniquely covered. The cover consists of the principal upsets of a partial order obtained by orienting the distance-two graph of G along a proper colouring; its members are stable as soon as the odd girth of G is large compared with the number of colours. Graphs of large girth, bounded degree and large chromatic number supply the hypothesis.

1 Introduction

All graphs are finite and simple. A set of vertices is *stable* if no two of its elements are adjacent. A *stable-set cover* of a graph G is a family $\mathcal{A}$ of stable subsets of $V(G)$ with $\bigcup \mathcal{A} = V(G)$; the members need not be disjoint. For a set $P \subseteq V(G)$ and $v \in P$, a member $X \in \mathcal{A}$ is *private for v on P* if $X \cap P = \{v\}$, and we say that P has the *private-cover property* (with respect to $\mathcal{A}$) if every $v \in P$ has a private member on P .

If $\mathcal{A}$ is a partition of $V(G)$ into stable sets, i.e. a proper colouring, then an induced path has the private-cover property exactly when its vertices receive pairwise distinct colours, that is, when it is a *rainbow* induced path. The main theorem of the ninth paper of the series of Scott and Seymour on induced subgraphs of graphs with large chromatic number [1] is that for every s , every triangle-free graph of sufficiently large chromatic number contains, for every proper colouring, an s -vertex rainbow induced path. In the conclusion of that paper they ask whether the partition hypothesis can be dropped:

Question 1.1 ([1, Conclusion]). Let G be a triangle-free graph with very large chromatic number, and let $\mathcal{A}$ be a set of stable subsets (not necessarily pairwise disjoint) of $V(G)$ with union $V(G)$. Does there necessarily exist an s -vertex induced path P of G such that for each $v \in V(P)$, some $X \in \mathcal{A}$ satisfies $X \cap V(P) = \{v\}$?

The natural reading is: for each fixed s , is there a threshold on $\chi(G)$ beyond which such a path always exists? We show that the answer is no for every $s \geq 3$, and that the failure is as strong as possible.

Theorem 1.2 (Main theorem). *For every integer $s \geq 3$ and every C , there is a finite connected triangle-free graph G with $\chi(G) > C$ and a family $\mathcal{A}$ of stable subsets of $V(G)$ with union $V(G)$, such that no s -vertex induced path P of G satisfies*

$$\forall v \in V(P) \exists X \in \mathcal{A} : X \cap V(P) = \{v\}.$$

In fact one G and one $\mathcal{A}$ serve for all $s \geq 3$ simultaneously: no path of G with at least three vertices, induced or not, has the private-cover property.

Corollary 1.3. *Question 1.1 has a negative answer for every $s \geq 3$. For $s \in \{1, 2\}$ the answer is trivially positive: every vertex lies in some member of $\mathcal{A}$, and for an edge uv every member containing u excludes v and vice versa. Thus $s = 3$ is exactly the threshold.*

Idea of proof. Two independent ingredients. (i) A deterministic obstruction (Lemma 2.1). Let $D(G)$ be the graph on $V(G)$ in which two distinct vertices are adjacent when they have a common neighbour in G . Properly colour $D(G)$ with q colours, orient each edge of $D(G)$ from the smaller to the larger colour, and let $\preceq$ be the resulting partial order. Take $\mathcal{A}$ to be the family of principal upsets $A_x = \{y : x \preceq y\}$. If the odd girth of G exceeds $4q - 3$, each A_x is stable: two adjacent vertices $y, z \in A_x$ would close an odd walk of length at most $4q - 3$ through x . Whenever u, w have a common neighbour they are comparable, say $u \preceq w$, and then every member containing u contains w ; so on any path $p_1 p_2 p_3 \dots$ one of p_1, p_3 has no private member. (ii) Graphs satisfying the hypothesis (Lemma 3.1). A random graph $G(n, d/n)$, after deleting the vertices of degree above $2d$ and one vertex from each cycle of length at most $20d^2$, has maximum degree at most $2d$, girth above $20d^2$ and chromatic number above $d/(14 \log d)$. Its distance-two graph has maximum degree below $4d^2$, so $q \leq 4d^2$ colours suffice and $20d^2 > 4q - 3$. [full proof]

2 The deterministic obstruction

For a graph G let $D(G)$ be the graph with vertex set $V(G)$ in which two distinct vertices are adjacent if and only if they have a common neighbour in G . The *odd girth* of G is the length of a shortest odd cycle (∞ if G is bipartite).

Lemma 2.1 (Nested-upset covers). *Suppose that $D(G)$ has a proper colouring with q colours and that the odd girth of G is greater than $4q - 3$. Then G has a stable-set cover $\mathcal{A}$ with the following property: whenever u, w have a common neighbour in G , one of them, say u , satisfies*

$$u \in X \in \mathcal{A} \implies w \in X.$$

Consequently no path in G with at least three vertices has the private-cover property with respect to $\mathcal{A}$.

Idea of proof. Orient the edges of $D(G)$ towards the larger colour; colours strictly increase along directed paths, so the reflexive transitive closure $\preceq$ is a partial order and directed paths have at most $q - 1$ edges. Each edge of $D(G)$ expands to a walk of length 2 in G , so two adjacent vertices of a principal upset A_x would give an odd closed walk of length at most $2(q - 1) + 2(q - 1) + 1 = 4q - 3$ in G , hence an odd cycle of at most that length. The domination property is transitivity of $\preceq$, and on a path $p_1 p_2 p_3 \dots$ the vertices p_1, p_3 share the neighbour p_2 . [full proof]

3 Graphs with large girth, bounded degree and large chromatic number

Lemma 3.1. *For all sufficiently large integers d there is a finite connected graph G with*

$$\Delta(G) \leq 2d, \quad \text{girth}(G) > 20d^2, \quad \chi(G) > \frac{d}{14 \log d}.$$

Idea of proof. This is Erdős’s argument [3] with an added degree cap. In $R \sim G(n, p)$ with $p = d/n$, $g = 20d^2$ and $n \geq 16gd^9$: the expected number of cycles of length at most g is at most $gd^g \leq n/16$; the expected number of vertices of degree above $2d$ is at most $(e/4)^d n \leq n/16$; and a union bound over r -sets with $r = \lceil 6n \log d/d \rceil$ gives $\Pr(\alpha(R) \geq r) < 1/4$. So with positive probability fewer than $n/4$ short cycles, fewer than $n/4$ high-degree vertices, and $\alpha(R) < r$. Deleting one vertex per short cycle and all high-degree vertices leaves more than $n/2$ vertices, and $\chi \geq |V|/\alpha$ gives the bound. [\[full proof\]](#)

Proof of Theorem 1.2. Let d be large enough for Lemma 3.1 and let G be the graph it provides. Since $\text{girth}(G) > 20d^2 > 3$, G is triangle-free, and $\chi(G) > d/(14 \log d)$, which tends to infinity with d ; choose d so that this exceeds C .

Two distinct vertices u, w are adjacent in $D(G)$ only if w is a neighbour of a neighbour of u , so

$$\Delta(D(G)) \leq \sum_{v \in N(u)} (\deg v - 1) \leq \Delta(G)(\Delta(G) - 1) \leq 2d(2d - 1),$$

and a greedy colouring shows that $D(G)$ has a proper colouring with $q \leq 2d(2d - 1) + 1 \leq 4d^2$ colours. On the other hand

$$\text{odd-girth}(G) \geq \text{girth}(G) > 20d^2 \geq 16d^2 > 4q - 3.$$

Lemma 2.1 therefore gives a stable-set cover $\mathcal{A}$ of $V(G)$ under which no path with at least three vertices, in particular no s -vertex induced path for any $s \geq 3$, has the private-cover property. $\square$

Proof of Corollary 1.3. The negative answer for $s \geq 3$ is Theorem 1.2. For $s = 1$, a vertex v lies in some $X \in \mathcal{A}$ and $X \cap \{v\} = \{v\}$. For $s = 2$, if $uv \in E(G)$ and $X \in \mathcal{A}$ contains u then $v \notin X$ since X is stable, so $X \cap \{u, v\} = \{u\}$; symmetrically for v . $\square$

4 Remarks

Remark 4.1 (Interpretation). The referee confirmed the wording of Question 1.1 against the TeX source of [1], including the clause “(not necessarily pairwise disjoint)”, which is the whole content of the question: the partition case is the paper’s main theorem and is true. The cover $\mathcal{A}$ of Lemma 2.1 is far from a partition (its members are nested along chains of $\preceq$ of length up to q), so Theorem 1.2 is consistent with that theorem and answers precisely the question asked.

Remark 4.2 (Strength of the refutation). A single cover blocks every path with at least three vertices, induced or not, in graphs of arbitrarily large girth and chromatic number. So neither fixing s (beyond $s \leq 2$), nor imposing a girth condition on G , rescues the question. What remains open are restricted variants: covers of bounded multiplicity (each vertex in a bounded number of members), or covers with a bounded nesting depth.

Remark 4.3 (Existential construction). The graphs of Lemma 3.1 come from the probabilistic method and are astronomically large ($n \geq 16 \cdot 20d^2 \cdot d^{20d^2}$); no explicit family is given. Any triangle-free graphs of large chromatic number, bounded maximum degree Δ and odd girth above $4\Delta(\Delta - 1) + 1$ would do equally well in Lemma 2.1.

Remark 4.4 (Computational checks). The referee implemented [Lemma 2.1](#) exactly as stated (DSATUR colouring of $D(G)$, colour-increasing orientation, principal upsets) and checked it on eight graphs satisfying the odd-girth hypothesis, among them $C_{13}, C_{17}, C_{21}, C_{25}$, a 100-vertex subdivision of the Petersen graph and a random 40-vertex tree: every A_x was stable, the union was $V(G)$, the domination property held on every edge of $D(G)$, and every induced path with 3 to 6 vertices failed the private-cover property, while all paths with 1 or 2 vertices passed. A control run with a proper-colouring partition produced rainbow induced paths wherever $\chi \geq 3$. The inequalities of [Lemma 3.1](#) were checked numerically for $d \in \{9, 12, 20, 50, 100, 1000\}$. See [Appendix B](#).

Remark 4.5 (Related work and novelty). The proper-colouring special case is the subject of the rainbow-path conjecture of Aravind; Duron and Raymond [2] recently improved the length of the guaranteed rainbow induced path in triangle-free graphs, but do not consider covers by overlapping stable sets. The referee's search found no prior resolution of [Question 1.1](#) in either direction; the construction is simple enough that independent earlier discovery is conceivable, and a human priority check is advised.

A Full proofs

Lemma 2.1 (Nested-upset covers). *Suppose that $D(G)$ has a proper colouring with q colours and that the odd girth of G is greater than $4q - 3$. Then G has a stable-set cover $\mathcal{A}$ with the following property: whenever u, w have a common neighbour in G , one of them, say u , satisfies*

$$u \in X \in \mathcal{A} \implies w \in X.$$

Consequently no path in G with at least three vertices has the private-cover property with respect to $\mathcal{A}$.

Proof. Fix a proper colouring $\varphi: V(G) \rightarrow \{1, \dots, q\}$ of $D(G)$. Orient every edge uw of $D(G)$ from the endpoint of smaller colour to the endpoint of larger colour; since φ is proper the colours differ, so this is well defined. Let $\preceq$ be the reflexive transitive closure of this orientation. Because φ strictly increases along every directed edge, the orientation is acyclic, $\preceq$ is a partial order, and every directed path in $D(G)$ has at most $q - 1$ edges (its vertices carry strictly increasing colours from $\{1, \dots, q\}$).

For $x \in V(G)$ let $A_x = \{y \in V(G) : x \preceq y\}$ be the principal upset of x .

Each A_x is stable in G . Suppose $y, z \in A_x$ with $yz \in E(G)$; in particular $y \neq z$. There are directed paths in $D(G)$ from x to y and from x to z , of lengths $a, b \leq q - 1$ respectively. Every edge of $D(G)$ joins two vertices with a common neighbour in G and can therefore be replaced by a walk of length 2 in G ; this yields walks in G from x to y and from x to z of lengths $2a$ and $2b$. Traverse the first walk backwards from y to x , then the second from x to z , then the edge zy . This is a closed walk in G of odd length

$$2a + 2b + 1 \leq 4q - 3.$$

Every closed walk of odd length contains an odd cycle of at most the same length, contradicting the hypothesis that the odd girth of G exceeds $4q - 3$. (The degenerate case $a = b = 0$ would force $y = z = x$, contradicting $y \neq z$.) Hence A_x is stable.

$\mathcal{A} = \{A_x : x \in V(G)\}$ is a cover. $x \in A_x$ by reflexivity.

Domination. Let u, w have a common neighbour in G . Then $uw \in E(D(G))$ and the orientation contains $u \rightarrow w$ or $w \rightarrow u$; say $u \rightarrow w$, so $u \preceq w$. If $u \in A_x$ then $x \preceq u \preceq w$, so $w \in A_x$. As every member of $\mathcal{A}$ is some A_x , every member containing u contains w .

No path with at least three vertices has the private-cover property. Let $P = p_1 p_2 \dots p_s$ with $s \geq 3$ be a path in G . The distinct vertices p_1, p_3 have the common neighbour p_2 . If $p_1 \preceq p_3$, every member of $\mathcal{A}$ containing p_1 also contains p_3 , so no $X \in \mathcal{A}$ satisfies $X \cap V(P) = \{p_1\}$. If $p_3 \preceq p_1$, the same holds for p_3 . Either way P fails the private-cover property. The argument does not use that P is induced. $\square$

Lemma 3.1. *For all sufficiently large integers d there is a finite connected graph G with*

$$\Delta(G) \leq 2d, \quad \text{girth}(G) > 20d^2, \quad \chi(G) > \frac{d}{14 \log d}.$$

Proof. Fix a sufficiently large integer d , put $g = 20d^2$, and choose n sufficiently large, in particular $n \geq 16gd^9$. Let $R \sim G(n, p)$ with $p = d/n$. We show that three events hold simultaneously with positive probability.

Few short cycles. Let Y be the number of cycles of R of length between 3 and g . For each ℓ the number of potential ℓ -cycles is $(n)_\ell / (2\ell)$, where $(n)_\ell = n(n-1)\cdots(n-\ell+1) \leq n^\ell$, so

$$\mathbb{E}[\#C_\ell] = \frac{(n)_\ell}{2\ell} p^\ell \leq \frac{d^\ell}{2\ell} \leq d^\ell, \quad \mathbb{E}Y \leq \sum_{\ell=3}^g d^\ell \leq gd^g \leq \frac{n}{16}.$$

By Markov's inequality, $\Pr(Y \geq n/4) \leq 1/4$.

Few vertices of degree greater than $2d$. Let Z be the number of vertices of R of degree greater than $2d$. The degree of a fixed vertex is $\text{Bin}(n-1, d/n)$, which is stochastically dominated by $\text{Bin}(n, d/n)$, a binomial variable of mean exactly d . The standard Chernoff bound $\Pr(X \geq 2\mu) \leq (e/4)^\mu$ for a binomial X of mean μ gives

$$\Pr(\deg_R(v) > 2d) \leq \left(\frac{e}{4}\right)^d \leq \frac{1}{16}$$

for all sufficiently large d (indeed for $d \geq 9$). Hence $\mathbb{E}Z \leq n/16$ and, by Markov's inequality, $\Pr(Z \geq n/4) \leq 1/4$.

Small independence number. Put $r = \lceil 6n \log d/d \rceil$. A union bound over all r -subsets, together with $\binom{n}{r} \leq (en/r)^r$ and $1-p \leq e^{-p}$, gives

$$\Pr(\alpha(R) \geq r) \leq \binom{n}{r} (1-p)^{\binom{r}{2}} \leq \exp\left(r \log \frac{en}{r} - \frac{p r(r-1)}{2}\right).$$

For sufficiently large d and n ,

$$r \leq \frac{7n \log d}{d}, \quad r-1 \geq \frac{5n \log d}{d}, \quad \log \frac{en}{r} \leq \log d,$$

the last because $en/r \leq ed/(6 \log d) \leq d$ once $6 \log d \geq e$. Hence the exponent is at most

$$\frac{7n(\log d)^2}{d} - \frac{d}{n} \cdot \frac{1}{2} \cdot \frac{6n \log d}{d} \cdot \frac{5n \log d}{d} = \frac{7n(\log d)^2}{d} - \frac{15n(\log d)^2}{d} = -\frac{8n(\log d)^2}{d},$$

which tends to $-\infty$ as $n \rightarrow \infty$. So, after increasing n if necessary, $\Pr(\alpha(R) \geq r) < 1/4$.

Conclusion. Since $1/4 + 1/4 + 1/4 < 1$, with positive probability $Y < n/4$, $Z < n/4$ and $\alpha(R) < r$; fix such an R . Delete every vertex of degree greater than $2d$, and delete one vertex from every cycle of length at most g . At most $Y + Z < n/2$ vertices are deleted; let G be the resulting graph. Then $|V(G)| > n/2$, every vertex of G has degree at most $2d$ (degrees do not increase under deletion), and G has no cycle of length at most g , i.e. $\text{girth}(G) > g = 20d^2$. The independence number is monotone under vertex deletion, so

$$\alpha(G) \leq \alpha(R) < r \leq \frac{7n \log d}{d}, \quad \text{hence} \quad \chi(G) \geq \frac{|V(G)|}{\alpha(G)} > \frac{n/2}{7n \log d/d} = \frac{d}{14 \log d}.$$

Finally, the chromatic number of a graph is the maximum over its components; replacing G by a component of maximum chromatic number keeps all three properties and makes G connected. $\square$

B Report of the LLM referee (verbatim)

The following is the unedited report of the adversarial referee agent (claude-fable-5, 2026-09-02) on the *original* writeup. Its step numbers refer to that writeup, not to this note (the writeup’s Lemma 1 and Lemma 2 are [Lemma 2.1](#) and [Lemma 3.1](#) here). The scripts it mentions are in `verification/scripts/1702.01094__01/`. Treat it as machine output, not as an authority.

B.1 Interpretation

The catalog statement was checked against the actual arXiv source. I downloaded the TeX source of arXiv:1702.01094 (Scott & Seymour, “Induced subgraphs of graphs with large chromatic number. IX. Rainbow paths”, v2) and located the question verbatim in the Conclusion (source lines 249-253):

“Let us also mention another question to which we do not know the answer. Let G be a triangle-free graph with very large chromatic number, and let $\mathcal{A}$ be a set of stable subsets (not necessarily pairwise disjoint) of $V(G)$ with union $V(G)$. Does there necessarily exist an s -vertex induced path P of G such that for each $v \in V(P)$, some $X \in \mathcal{A}$ satisfies $X \cap V(P) = \{v\}$?”

The extraction is word-for-word identical to the paper; no hypothesis was dropped. The natural quantification (for each s , does some threshold on χ suffice?) is exactly what the writeup negates: for every $s \geq 3$ and every C it exhibits a triangle-free (indeed huge-girth) connected G with $\chi(G) > C$ and a stable-set cover $\mathcal{A}$ admitting no such path of any length ≥ 3 .

This is **not** interpretation gaming. The parenthetical “(not necessarily pairwise disjoint)” is the authors’ own emphasis: the entire content of the question is whether the rainbow-path theorem (the partition/proper-coloring case, which is the paper’s main theorem and is true) survives when overlap is allowed. The counterexample exploits precisely the overlap the authors explicitly permitted, and its cover is far from a partition, so it is consistent with the main theorem. A reasonable author of the source paper would consider their question answered (in the negative). Interpretation: fair and intended.

B.2 Step-by-step findings

step	label	note
1. Theorem statement (negation of the question for all $s \geq 3$, all C)	VALID	Correctly negates the question under its natural quantification; a single cover kills all $s \geq 3$ simultaneously.
2. Lemma 1 setup: $D(G)$ (common-neighbor graph), proper q -coloring φ , low-to-high-color orientation, $\preceq$ = reflexive transitive closure	VALID	Colors strictly increase along directed edges, so the orientation is acyclic, $\preceq$ is a partial order, and every directed path has $\leq q - 1$ edges ($\leq q$ vertices with strictly increasing colors).

step	label	note
3. Every principal upset $A_x = \{y : x \preceq y\}$ is stable in G	VALID	If $y, z \in A_x$ with $yz \in E(G)$: directed $D(G)$ -paths $x \rightarrow y$, $x \rightarrow z$ of lengths $a, b \leq q - 1$; each $D(G)$ -edge expands to a 2-edge G -walk; the closed walk $y \rightarrow x \rightarrow z \rightarrow y$ has odd length $2a + 2b + 1 \leq 4q - 3$; an odd closed walk contains an odd cycle of at most its length, contradicting odd girth $> 4q - 3$. Degenerate cases impossible ($a = b = 0$ forces $y = z = x$, but $yz \in E(G)$ needs $y \neq z$). Verified computationally on 8 graphs (see below).
4. $\mathcal{A} = \{A_x\}$ covers $V(G)$	VALID	$x \in A_x$ by reflexivity. Verified computationally.
5. Domination: if u, w share a neighbor and $u \rightarrow w$, every member of $\mathcal{A}$ containing u contains w	VALID	$u \in A_x \Rightarrow x \preceq u \preceq w \Rightarrow w \in A_x$; $\mathcal{A}$ consists only of principal upsets, so the quantification over members is complete. Verified computationally on every $D(G)$ -edge of every test graph.
6. No path $p_1 \cdots p_s$ ($s \geq 3$) has the private-cover property	VALID	p_1, p_3 share the neighbor p_2 and are distinct; whichever of them is dominated, say p_1 , has $p_3 \in X$ for every $X \ni p_1$, so no X has $X \cap V(P) = \{p_1\}$. (The argument does not even need the path induced.) Verified by exhaustive enumeration of all induced paths, $s = 3..6$, on all test graphs: zero surviving paths.
7. Lemma 2, few short cycles: $\mathbb{E}[\#C_\ell] = \frac{(n)_\ell}{2^\ell} p^\ell \leq \frac{d^\ell}{2^\ell}$, $\mathbb{E}Y \leq gd^g \leq n/16$, Markov gives $\Pr(Y \geq n/4) \leq 1/4$	VALID	Re-derived: $(n)_\ell p^\ell \leq d^\ell$; $\sum_{\ell=3}^g d^\ell \leq gd^g$; $n \geq 16gd^g$ by choice.
8. Few high-degree vertices: $\Pr(\deg > 2d) \leq (e/4)^d \leq 1/16$, $\Pr(Z \geq n/4) \leq 1/4$	VALID	$\deg \sim \text{Bin}(n-1, d/n) \preceq \text{Bin}(n, d/n)$ (mean exactly d); Chernoff $\Pr(X \geq 2\mu) \leq (e/4)^\mu$ is the standard $\delta = 1$ bound. $(e/4)^d \leq 1/16$ for $d \geq 9$ (checked numerically: $(e/4)^9 \approx 0.031$).

step	label	note
9. Small independence number: $r = \lceil 6n \log d/d \rceil$, exponent $\leq 7n(\log d)^2/d -$ $15n(\log d)^2/d =$ $-8n(\log d)^2/d$	VALID	$\binom{n}{r}(1-p)^{\binom{r}{2}} \leq \exp(r \log \frac{en}{r} - \frac{pr(r-1)}{2})$ is standard; the three intermediate bounds hold for large d, n ($\log \frac{en}{r} \leq \log \frac{ed}{6 \log d} \leq \log d$ needs only $6 \log d \geq e$, i.e. $d \geq 2$); the arithmetic $7 - \frac{1}{2} \cdot 6 \cdot 5 = -8$ re-derived and machine-checked, and the full exponent evaluated numerically for several (d, n) . $1/4 + 1/4 + 1/4 < 1$; at most $Y + Z < n/2$ deletions kill all cycles of length $\leq g$ and all degrees $> 2d$; α monotone under vertex deletion; $\chi \geq V /\alpha$; connectivity by taking a component of maximum chromatic number.
10. Union bound, deletion, and conclusion of Lemma 2: $ V(G) > n/2$, $\Delta(G) \leq 2d$, girth $> g = 20d^2$, $\alpha(G) < r$, $\chi(G) \geq \frac{n/2}{7n \log d/d} = \frac{d}{14 \log d}$	VALID	Common-neighbor count $\leq \sum_{u \in N(v)} (\deg u - 1) \leq \Delta(\Delta - 1)$; $2d(2d - 1) + 1 = 4d^2 - 2d + 1 \leq 4d^2$ for $d \geq 1$; girth $> 20 > 3$ also gives triangle-freeness. $d/(14 \log d) \rightarrow \infty$, so χ is unbounded.
11. Application: $\Delta(D(G)) \leq \Delta(G)(\Delta(G) - 1) \leq 2d(2d - 1)$, greedy $q \leq 4d^2$, odd girth $> 20d^2 > 16d^2 - 3 \geq 4q - 3$	VALID	$s = 1$: any vertex is in some cover member, whose intersection with $\{v\}$ is $\{v\}$. $s = 2$: stability separates the endpoints of an edge. Verified computationally (all 1- and 2-vertex induced paths pass on every test graph).
12. Small cases $s = 1, 2$ trivially positive	VALID	

No step is labeled GAP or ERROR.

B.3 Reference check

- **Source question** (arXiv:1702.01094, Conclusion): confirmed verbatim from the downloaded TeX source (arxiv.org/e-print/1702.01094, file paper.tex, lines 249-253). The extracted statement matches word for word, including “(not necessarily pairwise disjoint)”. Confirmed the paper’s main theorem is the proper-coloring (partition) case, so the counterexample does not contradict it.
- **External results used by the writeup**: none. The proof is self-contained, invoking only textbook facts (an odd closed walk contains an odd cycle; greedy coloring uses $\Delta + 1$ colors; Markov; the standard Chernoff bound $\Pr(X \geq 2\mu) \leq (e/4)^\mu$; $\binom{n}{r} \leq (en/r)^r$; $\chi \geq n/\alpha$). Each was re-derived above rather than taken on faith. Lemma 2 is a self-contained reproof of the

classical Erdős-type high-girth/high-chromatic existence with an added degree cap; no citation is needed or made.

- **Priority search:** web searches (September 2026) and the catalog’s own auto-review (May 2026) found no work addressing the general stable-set-cover formulation. The only related progress (Duron-Raymond, arXiv:2601.00602) targets the proper-coloring rainbow-path conjecture; I fetched its full text and it nowhere mentions the not-necessarily-disjoint cover question. I found no prior resolution, positive or negative.

B.4 Computational check

Scripts in `verification/scripts/1702.01094__01/` (run with Python 3.14 + networkx 3.x in a venv):

- `check_lemma1.py` implements Lemma 1 exactly as written ($D(G)$, DSATUR proper coloring, color-increasing orientation, principal upsets) and brute-forces its claims on eight graphs satisfying the hypothesis odd-girth $> 4q - 3$: odd cycles $C_{13}, C_{17}, C_{21}, C_{25}$ (all with $q = 3, 4q - 3 = 9$), P_{20}, C_{24} , the Petersen graph with every edge subdivided into a 7-edge path ($n = 100$, odd girth 35, $q = 3$), and a random 40-vertex tree ($q = 6, 4q - 3 = 21$). On every graph: all A_x stable (0 violations), union $= V(G)$, domination holds on every $D(G)$ -edge (up to 120 edges), **every** induced path with $s = 3, 4, 5, 6$ vertices fails the private-cover property (e.g. C_{25} : 25 induced paths per length, 0 pass; subdivided Petersen: 120/135/150 induced paths for $s = 3/4/5$, 0 pass), while all $s = 1, 2$ paths pass, matching the writeup’s trivial cases. A control run with a proper-coloring **partition** cover produced rainbow induced P_3 ’s wherever $\chi \geq 3$ (e.g. 3 on each odd cycle, 16 on subdivided Petersen), confirming consistency with the Scott-Seymour theorem.
- `check_lemma2_arith.py` machine-checks the arithmetic of Lemma 2 for $d \in \{9, 12, 20, 50, 100, 1000\}$: $(e/4)^d \leq 1/16$ (e.g. 3.09×10^{-2} at $d = 9$), $\sum_{\ell=3}^g d^\ell \leq gd^g$, the r -bounds $5n \log d/d \leq r - 1 < r \leq 7n \log d/d$, $\log(en/r) \leq \log d$, the direct exponent evaluation confirming $r \log \frac{en}{r} - \frac{pr(r-1)}{2} \leq -8n(\log d)^2/d$ (e.g. $-7.87 \times 10^{10} \leq -3.86 \times 10^{10}$ at $d = 9, n = 9 \times 10^9$), the identity $7 - \frac{1}{2} \cdot 6 \cdot 5 = -8$ in exact rationals, $4q - 3 < 20d^2$ for $q = 2d(2d - 1) + 1$, and χ -bound algebra $\frac{n/2}{7n \log d/d} = \frac{d}{14 \log d}$.

Both scripts print ALL ... CHECKS PASSED. Note the final counterexample graphs themselves ($n \geq 16 \cdot 20d^2 \cdot d^{20d^2}$) are astronomically large and existential, so they cannot be built; what the code verifies is the deterministic core (Lemma 1, the load-bearing novelty) end-to-end on real graphs, plus every inequality in the probabilistic part.

B.5 Caveats

1. **Existential, not explicit.** The high-chromatic counterexamples come from the probabilistic method; no explicit family is given. This does not affect correctness (the writeup says so itself).
2. **Priority not fully excluded.** I found no prior resolution in searches or in the catalog’s literature review, but a human check (e.g. asking Scott/Seymour or scanning all citing papers) is the only way to be certain the negative answer is new. The construction is simple enough that independent prior discovery is conceivable.
3. **Chernoff mean.** The degree is $\text{Bin}(n - 1, d/n)$ with mean slightly below d ; the writeup’s “stochastically dominated by a binomial of mean d ” ($\text{Bin}(n, d/n)$) is the correct and standard fix — cosmetic only.
4. **What survives of the question.** The counterexample cover has heavily nested members (chains of length up to $q \approx 4d^2$), and each vertex can lie in many members. Restricted variants — covers of bounded multiplicity, or with a bound on the number of members containing each vertex — are untouched and remain open, as does the partition case (true, by the paper’s main theorem).
5. **Strength of the refutation.** The failure is total: a single cover blocks all paths (induced or not) with ≥ 3 vertices, for graphs of arbitrarily large girth and chromatic number, so no weakening by fixing s small (beyond $s \leq 2$) or imposing girth conditions on G rescues the question.

6. **Quantifier reading.** “Very large chromatic number” is informal in the source; under any reasonable reading (threshold depending on s , or $\chi \rightarrow \infty$ with s fixed) the writeup’s theorem negates it for every $s \geq 3$.

B.6 Referee summary

I set out to find the error and could not. The deterministic heart of the disproof — Lemma 1’s construction of a stable-set cover by principal upsets of a color-increasing orientation of the distance-two graph, under the hypothesis odd-girth $> 4q - 3$ — is correct, and I verified it exhaustively by computer on eight graphs, including a 100-vertex subdivided Petersen graph: the cover is always stable, always covering, and never admits a privately-covered induced path on 3 or more vertices, exactly as claimed. The probabilistic Lemma 2 is a standard Erdős-type argument with generous slack; every inequality was re-derived by hand and machine-checked numerically. The question was confirmed verbatim against the arXiv TeX source, including the crucial “(not necessarily pairwise disjoint)” clause, so the interpretation is the authors’ intended one and the negative answer resolves it fairly — the counterexample cover is not a partition, so the paper’s main theorem is untouched. No prior resolution surfaced in literature searches. Verdict: CONFIRMED (subject to a human priority check).

References

- [1] A. Scott and P. Seymour, Induced subgraphs of graphs with large chromatic number. IX. Rainbow paths, arXiv:1702.01094 (2017).
- [2] J. Duron and J.-F. Raymond, Towards a conjecture on long induced rainbow paths in triangle-free graphs, arXiv:2601.00602 (2026).
- [3] P. Erdős, Graph theory and probability, *Canad. J. Math.* 11 (1959) 34–38.